

IMO 2026 — Problem 4

Solution by Claude Fable 5

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Source: `results/claude-fable-5/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

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Approaches tried

- **Cut parametrization** (`lemmas/cut-parametrization.md`): a cut through the vertex with angle A , with $\alpha = \angle BAP \in (0, A)$, produces children with angle multisets $\{B, \alpha, 180^\circ - B - \alpha\}$

and $\{C, A - \alpha, B + \alpha\}$. Verified numerically (`code/verify.py`, `geo_check`). This is the foundation of everything below.

- **Immediate-win analysis:** with generic angles, a single cut can put a multiple of θ into *both* children only if the split angle is a multiple of θ , or if the two supplementary angles at P are both multiples of θ — the latter forces $180^\circ \in \theta\mathbb{N}$. This suggested the dichotomy “ θ divides 180° or not”.
- **Mulan’s strategy for $\theta = 180^\circ/n$** (`approaches/mulan-strategy-divisor-theta.md`): chain lemma (angle $k\theta$ win in $\leq k - 1$ cuts) plus a double-threat cut through the largest angle making both angles at P multiples of θ . Complete; verified exactly for $n = 2, \dots, 8$ against random play.
- **Shan-Yu’s invariant for $180^\circ/\theta \notin \mathbb{Z}$** (`approaches/shanyu-invariant.md`): keep all three angles outside $\theta\mathbb{N}$; a 4-case algebraic check shows this is always possible. Complete; verified exactly for several such θ with adversarial cut values.
- **Dead ends:** order-type invariants for Shan-Yu (“all angles $< \theta$ ”, “all angles $> \theta$ ”, “all in $(\theta, 2\theta)$ ”) — only the first works and only for $\theta > 90^\circ$; all subsumed by the arithmetic invariant.

Current best

Answer: Mulan can guarantee victory exactly for the angles θ such that $180^\circ/\theta$ is an integer, i.e. $\theta = 180^\circ/n$ for some integer $n \geq 2$ (namely $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, \dots\}$). If $n\theta = 180^\circ$, Mulan wins within at most $n - 1$ cuts via a “double threat + chain” strategy; if 180° is not an integer multiple of θ , Shan-Yu survives forever by keeping all three angles outside $\{\theta, 2\theta, 3\theta, \dots\}$, which a short case analysis shows is always possible. Full proof below; all key computations sanity-checked exactly in `code/verify.py`.

Full proof

Throughout, we identify a triangle with the (unordered) triple of its angles, all measured in degrees; the triple consists of positive reals summing to 180° . Let

$$\theta\mathbb{N} := \{k\theta : k = 1, 2, 3, \dots\}$$

denote the set of positive integer multiples of θ . Note that only the finitely many elements of $\theta\mathbb{N} \cap (0^\circ, 180^\circ)$ can occur as angles of a triangle.

Answer. Mulan can guarantee victory in finitely many steps if and only if

$$\theta = \frac{180^\circ}{n} \quad \text{for some integer } n \geq 2,$$

equivalently, iff $180^\circ \in \theta\mathbb{N}$. (Since $0^\circ < \theta < 180^\circ$, the integer $n = 180^\circ/\theta$ automatically satisfies $n \geq 2$.)

Lemma 0 (Cut parametrization)

Let $\mathcal{T}$ be a triangle with angles A, B, C at vertices also named A, B, C . Mulan’s legal cuts through vertex A (i.e. with P in the interior of side BC) are parametrized bijectively by $\alpha := \angle BAP \in (0^\circ, A)$, and the resulting two triangles have angle multisets

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\}, \quad T_2 = \{C, A - \alpha, B + \alpha\}.$$

Proof. As P traverses the open segment BC , the ray AP sweeps the open angular sector at A between the rays AB and AC ; the map $P \mapsto \alpha = \angle BAP$ is a continuous strictly monotone bijection from the interior of BC onto $(0^\circ, A)$. Triangle ABP has angle B at B and α at A , hence angle $180^\circ - B - \alpha$ at P . Triangle APC has angle C at C and $A - \alpha$ at A , hence at P the angle

$$180^\circ - C - (A - \alpha) = (180^\circ - A - B - C) + B + \alpha = B + \alpha. \quad \blacksquare$$

Note for later use: the two angles at P , namely $180^\circ - B - \alpha$ (in T_1) and $B + \alpha$ (in T_2), are supplementary.

A **round** consists of: the stopping check, then (if the game did not stop) one cut by Mulan and one discard by Shan-Yu. “Mulan wins within m cuts” means: whatever Shan-Yu does, the stopping check succeeds after at most m cuts have been made.

Part 1: If $n\theta = 180^\circ$ for an integer $n \geq 2$, Mulan wins (within at most $n - 1$ cuts)

Here $\theta\mathbb{N} \cap (0^\circ, 180^\circ) = \{k\theta : 1 \leq k \leq n - 1\}$.

Lemma 1 (Chain lemma). *For any $\theta \in (0^\circ, 180^\circ)$: if at the start of a round the triangle has an angle equal to $k\theta$ for some integer $k \geq 1$, then Mulan can win within at most $k - 1$ further cuts.*

Proof. Induction on k .

Base $k = 1$: the triangle has an angle exactly θ , so the stopping check ends the game at once; Mulan has won with 0 further cuts.

Step $k \geq 2$: let $A = k\theta$ be such an angle (note $k\theta < 180^\circ$), and let B, C be the other two angles. Since $0^\circ < \theta < k\theta = A$, Lemma 0 allows Mulan to cut through vertex A with $\alpha = \theta$. The children are

$$T_1 = \{B, \theta, 180^\circ - B - \theta\}, \quad T_2 = \{C, (k - 1)\theta, B + \theta\}.$$

If Shan-Yu keeps T_1 , then the next round starts with a triangle having an angle exactly θ , and the game stops with Mulan’s victory: $1 \leq k - 1$ cut was used. If Shan-Yu keeps T_2 , it has the angle $(k - 1)\theta$, so by the induction hypothesis Mulan wins within $k - 2$ further cuts, for a total of at most $k - 1$. $\blacksquare$

Lemma 2 (Double threat). *Assume $n\theta = 180^\circ$, $n \geq 2$. If at the start of a round the triangle has no angle in $\theta\mathbb{N}$, then Mulan has a cut such that **both** children contain an angle of the form $k\theta$ with $1 \leq k \leq n - 1$.*

Proof. Let A be a largest angle of the triangle and B, C the other two, so $A \geq 60^\circ$. We claim there is an integer $k \geq 1$ with

$$C < k\theta < 180^\circ - B. \quad (*)$$

The interval $(C, 180^\circ - B)$ is open with length $180^\circ - B - C = A$.

Case $n \geq 3$: then $\theta \leq 60^\circ \leq A$, and $A \notin \theta\mathbb{N}$ (hypothesis) gives $A \neq \theta$, hence $A > \theta$. Let k be the least positive integer with $k\theta > C$ (it exists since $\theta > 0$). Since $C \notin \theta\mathbb{N}$ and $(k - 1)\theta \leq C$, we get $k\theta \leq C + \theta < C + A = 180^\circ - B$. Thus $(*)$ holds.

Case $n = 2$ (so $\theta = 90^\circ$): we show $B, C < 90^\circ$, whence $(*)$ holds with $k = 1$: indeed $C < 90^\circ$ and $180^\circ - B > 90^\circ$. To see $B, C < 90^\circ$: if $A < 90^\circ$ then all three angles are $< 90^\circ$; $A = 90^\circ$ is impossible because no angle lies in $\theta\mathbb{N} \ni 90^\circ$; and if $A > 90^\circ$ then $B + C = 180^\circ - A < 90^\circ$, so $B, C < 90^\circ$.

Fix such k , and note $k\theta < 180^\circ - B < 180^\circ = n\theta$ forces $1 \leq k \leq n-1$. Mulan cuts through vertex A with

$$\alpha := 180^\circ - B - k\theta.$$

By (*), $0 < \alpha < 180^\circ - B - C = A$, so the cut is legal by Lemma 0. The children are

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\} = \{B, 180^\circ - B - k\theta, k\theta\},$$

$$T_2 = \{C, A - \alpha, B + \alpha\} = \{C, k\theta - C, 180^\circ - k\theta\} = \{C, k\theta - C, (n-k)\theta\},$$

where we used $A - \alpha = A - 180^\circ + B + k\theta = k\theta - C$ and $180^\circ - k\theta = (n-k)\theta$ with $1 \leq n-k \leq n-1$. Thus T_1 contains $k\theta$ and T_2 contains $(n-k)\theta$, both in $\{\theta, \dots, (n-1)\theta\}$. ■

Mulan's complete strategy for $\theta = 180^\circ/n$. At the start of the game (round 1), either the triangle already has an angle $k\theta$ for some $1 \leq k \leq n-1$ — then by Lemma 1 Mulan wins within $k-1 \leq n-2$ cuts — or it has no angle in $\theta\mathbb{N}$. In the latter case Mulan plays the cut of Lemma 2; whichever child Shan-Yu keeps, the next round starts with a triangle containing an angle $k'\theta$ with $1 \leq k' \leq n-1$, and by Lemma 1 Mulan wins within $k'-1 \leq n-2$ further cuts. In total Mulan wins within at most $n-1$ cuts, no matter how Shan-Yu chooses the initial triangle and his discards. This proves Part 1.

Part 2: If $180^\circ \notin \theta\mathbb{N}$, Shan-Yu can avoid defeat forever

Call a triangle **safe** if none of its three angles belongs to $\theta\mathbb{N}$. Since $\theta \in \theta\mathbb{N}$, a safe triangle never triggers the stopping check.

Lemma 3 (Existence of a safe start). *A safe triangle exists.*

Proof. The set $F := \theta\mathbb{N} \cap (0^\circ, 180^\circ)$ is finite. Pick any $A \in (0^\circ, 180^\circ) \setminus F$ (possible: F is finite, the interval is infinite). Now pick $B \in (0^\circ, 180^\circ - A)$ with $B \notin F$ and $180^\circ - A - B \notin F$: this excludes only finitely many values of B from an interval of positive length, so such B exists. Set $C = 180^\circ - A - B > 0$. Then (A, B, C) is safe. ■

Lemma 4 (Closure). *Assume $180^\circ \notin \theta\mathbb{N}$. If the current triangle is safe, then for every legal cut of Mulan, at least one of the two children is safe.*

Proof. Suppose Mulan cuts through the vertex whose angle is A ; let B, C be the other two angles and $\alpha \in (0^\circ, A)$ the cut parameter. By Lemma 0 the children are

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\}, \quad T_2 = \{C, A - \alpha, B + \alpha\}.$$

By safety, $B \notin \theta\mathbb{N}$ and $C \notin \theta\mathbb{N}$. Hence: - if T_1 is unsafe, then $\alpha \in \theta\mathbb{N}$ or $180^\circ - B - \alpha \in \theta\mathbb{N}$; - if T_2 is unsafe, then $A - \alpha \in \theta\mathbb{N}$ or $B + \alpha \in \theta\mathbb{N}$.

Assume for contradiction both children are unsafe. There are four cases, with $k, m \geq 1$ integers:

1. $\alpha = k\theta$ and $A - \alpha = m\theta$. Then $A = (k+m)\theta \in \theta\mathbb{N}$, contradicting safety of the current triangle.
2. $\alpha = k\theta$ and $B + \alpha = m\theta$. Then $B = (m-k)\theta$; since $B > 0$, $m-k \geq 1$, so $B \in \theta\mathbb{N}$ — contradiction.
3. $180^\circ - B - \alpha = k\theta$ and $A - \alpha = m\theta$. Subtracting: $180^\circ - A - B = (k-m)\theta$, i.e. $C = (k-m)\theta$; since $C > 0$, $k-m \geq 1$, so $C \in \theta\mathbb{N}$ — contradiction.
4. $180^\circ - B - \alpha = k\theta$ and $B + \alpha = m\theta$. Adding: $180^\circ = (k+m)\theta \in \theta\mathbb{N}$ — contradicting the hypothesis of Part 2.

All cases are impossible, so at least one child is safe. ■

Shan-Yu's complete strategy for $180^\circ \notin \theta\mathbb{N}$. He creates a safe initial triangle (Lemma 3). Whenever Mulan cuts, he keeps a safe child, which exists by Lemma 4. By induction, at the start of every round the triangle is safe, hence has no angle equal to θ , hence the stopping check never succeeds. Therefore no strategy of Mulan wins in finitely many steps: against Shan-Yu's strategy the game runs forever. This proves Part 2.

Conclusion

Combining Parts 1 and 2: Mulan can guarantee victory in finitely many steps **iff** 180° is an integer multiple of θ , i.e.

$$\theta = \frac{180^\circ}{n} \text{ for some integer } n \geq 2$$

(equivalently $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, 180^\circ/7, \dots\}$), and when she can, she needs at most $n - 1$ cuts. ■